KEYENCE

Application Guide for All Industries

Resin/ Plastic Laser Marking Techniques

- Resin marking/processing examples
- Principle of resin colour development
- Characteristics of each wavelength
- Absorption ratio difference among materials

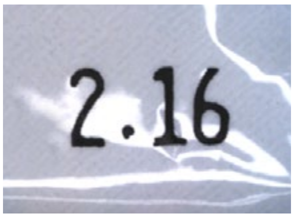
Relationship between resin products and laser markers

“Direct marking” on resin products

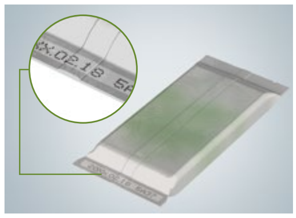
“Direct marking” is a method of indicating product IDs or time limits by marking data directly on products instead of using labels or name plates. Although a variety of marking methods exist, the advantages of “direct marking”, with lasers, has been attracting attention due to the need for reliable manufacturing history management.

Typical marking examples

Stamp



Industrial inkjet printer



Labeller

Laser Marker

Characteristics of each marking method

	Stamp	Label	Inkjet printer	Laser Marker
No fading	Normal	Good	Good	Excellent
Permanent marking (No coming off)	Poor	Normal	Normal	Excellent
Unaffected by materials	Good	Good	Good	Normal
Less waste	Good	Normal	Normal	Excellent
Marking on uneven surfaces	Normal	Normal	Good	Good
Flexibility to changes in marking data	Poor	Good	Excellent	Excellent
Initial cost	Excellent	Good	Normal	Normal
Running cost	Normal	Poor	Normal	Excellent

Advantages of laser markers

Permanent marking

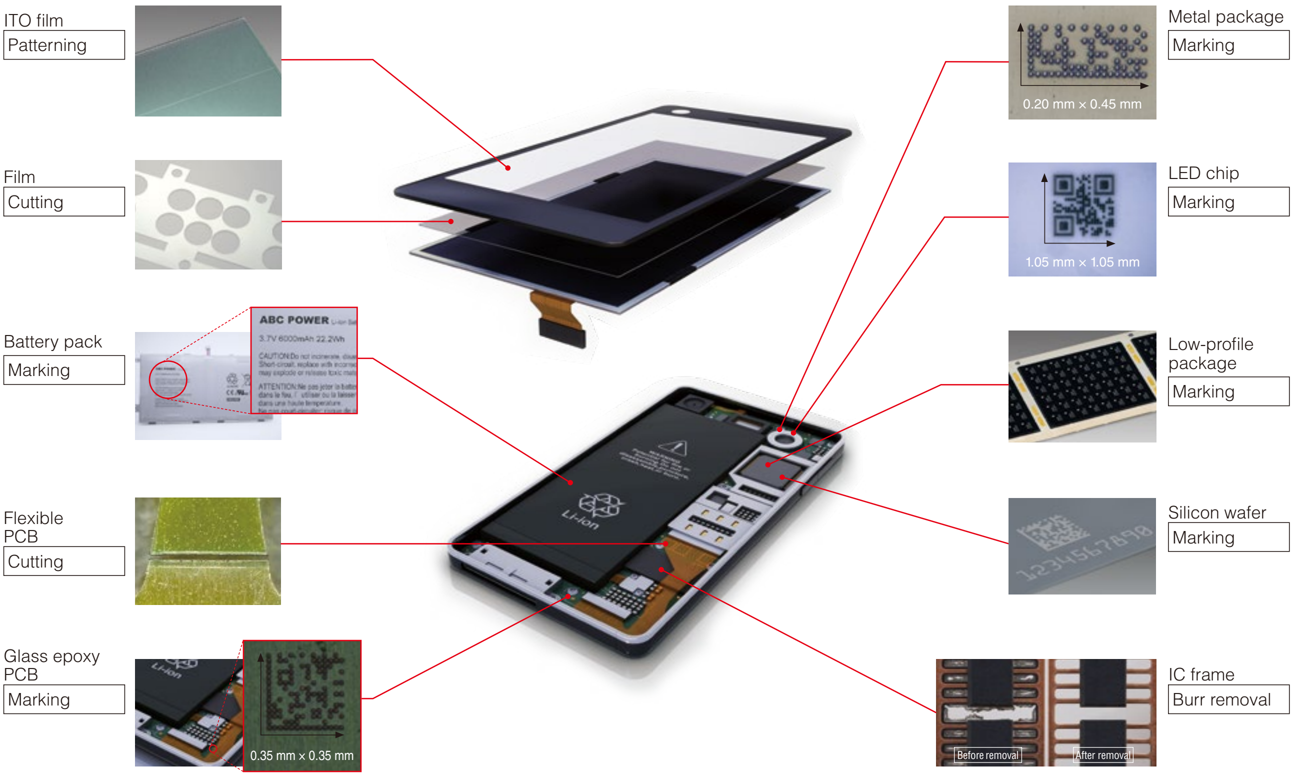
Low running cost

High marking quality

Less marking errors

Marking/processing on resin products

Laser markers have been introduced not only for marking applications but also for resin processing such as cutting and burr removal. In smartphone manufacturing processes, for example, laser markers are used for various applications other than marking.



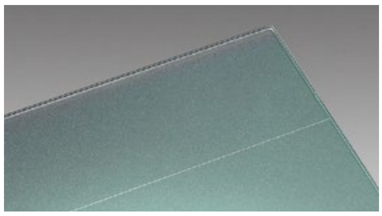
Typical laser processing applications

[Film cutting]

Laser markers have been widely introduced for film and sheet processing applications. When compared with stamping parts out using a mould, lasers have some clear advantages. Laser markers allow users to easily change between designs while reducing running costs since the non-contact method does not rely on moulds that can wear down over time.

[Burr removal from IC chips]

Flash burrs on the leads can be removed with a laser marker. Since the laser beam accurately scans the outline of the IC only, burrs can be removed without any damage to the package inside. The non-contact operation ensures high quality processing without any damage or deformation of the lead frame itself.



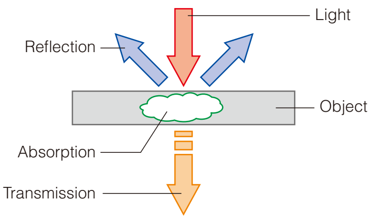
[ITO film patterning]

A laser can be used for the patterning of a surface layer. Although wet etching using chemicals has been the mainstream method so far, dry etching using a laser marker improves processing accuracy while reducing the environmental burden.

Laser markers can be used for not only marking but also for non-contact processing.

Mechanism of resin marking/processing

Reflection, Absorption, and Transmission



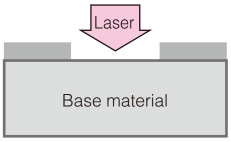
When light is received, all objects experience “reflection”, “absorption”, and “transmission”. This relationship is a very important element for laser marking and laser processing. The ratio of “reflection”, “absorption”, and “transmission” of the received light, form the following relationship.

Reflectivity + absorption rate + transmittance = 1

Without increasing the temperature of the object, processing becomes more difficult as “reflection” and “transmission” increases. Processing efficiency becomes better as “absorption” increases.

Major types of marking/processing

1: Paint peeling

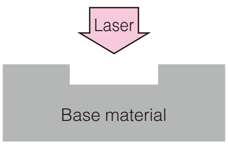


Peel the paint or printing on the target surface to bring out the contrast with the colour of the base material.

(Example) Automobile instrument panel switch

When the design is changed, conventional methods using printing or stamps required the printing plate to be changed. With a laser marker, you can handle it flexibly by just changing the program.

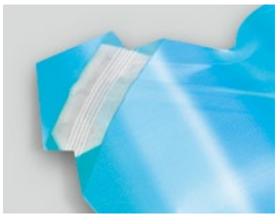
2: Surface layer peeling



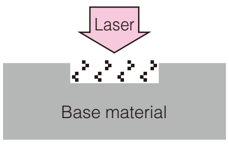
Remove/engrave the surface layer with a laser.

(Example) Half cut

Use a laser marker to process a cutting section. A cutter was used in the conventional method; however, there were problems such as difficult adjustment and time-consuming changeover between product types. Moreover, the method incurred costs for replacing the blade and there was a risk of the blade being left in the product.



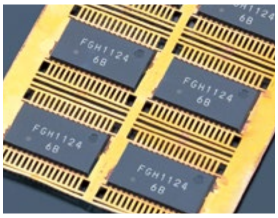
3: Colour development



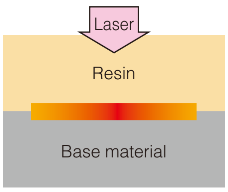
Irradiate a resin target with a laser to develop a colour in the target itself.

(Example) Wide-area marking on LSI

By using laser radiation to develop a colour instead of engraving the resin, the damage of marking the target can be minimised.



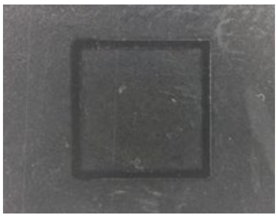
4: Welding



Use the heat of laser radiation to weld and join resin parts.

(Example) Welding of transparent resin and coloured resin

Traditional welding can cause vibration which may affect the product or produce burrs. Since laser welding does not make contact with products, it won’t damage them or produce burrs.



Continuous wave (CW) oscillation and pulsed oscillation

There are two methods of laser oscillation as follows.

Continuous wave (CW) oscillation

Emit a laser continuously to apply constant heat.

[A laser beam is emitted constantly.]

● Major applications: Welding, marking on IC packages

Pulsed oscillation

Change the frequency repeatedly to adjust pulse energy.

[A laser beam is stored and then emitted.]

● Major applications: Peeling, colour development

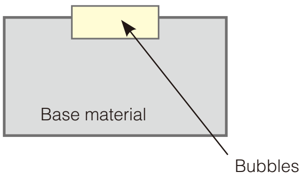
⇨ KEYENCE laser markers allow switching between continuous wave (CW) oscillation and pulsed oscillation, so that they can be used for every resin material.

Mechanism of resin colour development

The principle of colour development is largely divided into four steps.

1 Foaming	2 Condensing	3 Carbonisation	4 Chemical change

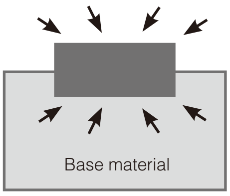
1: Foaming



When the base material is irradiated with a laser, gas bubbles are generated inside the material due to the thermal effect of the radiation. Gasified, evaporated bubbles are contained in the surface layer of the base material and create a whitish swelling. When the base material has a dark colour, the swelling is highly visible and typically paler than it was before marking.

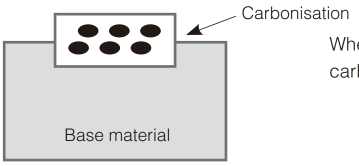
(Example) Base material colour: Black → Changes to Grey
Red → Changes to Pink

2: Condensing



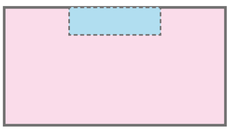
When the base material absorbs the laser energy, the thermal effect increases the molecular density. The molecules are condensed and the colour changes to a darker colour.

3: Carbonisation



When the area continuously receives high energy, macromolecules of the element around the base material are carbonised and turn black.

4: Chemical change

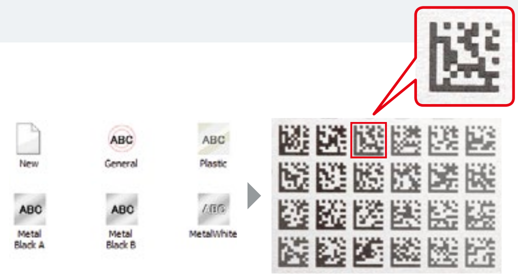


The “pigment” elements in the base material always contain metallic ions. The laser radiation changes the crystal structure of the ions and the hydration level in the crystal. Consequently, the composition of the element itself changes chemically, resulting in colour development due to the increased intensity of the pigment.

How can I find the optimal marking conditions for each material?

The “Sample Marking” function of KEYENCE laser markers can be used to find the best marking conditions for a particular material.

The colour of the contrast marking on resin targets greatly depends on the marking conditions of the laser marker.
With the “Sample Marking” function, all you have to do is to select the material, and the software runs the conditions automatically.
You can find the optimal marking conditions from the list of marking results. Setting the marking conditions, which normally requires some experience, can be completed by anyone within a short time.



Select a material.

The optimal conditions can be found immediately.

Influence on materials by wavelength

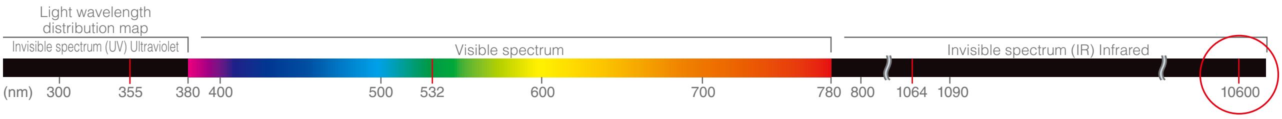
● Laser marker classification by wavelength

- CO_2 laser marker — **Wavelength: 10600 nm**
- YVO_4 laser marker, Fibre laser marker* — **Wavelength: 1064 nm*** (Standard wavelength)
- Green laser marker — **Wavelength: 532 nm** (SHG wavelength)

Industrial laser markers are broadly divided into four types based on the wavelength of the laser beam: CO_2 laser markers, YVO_4 laser markers, fibre laser markers, and SHG (green) laser markers.
In general, the shorter the wavelength of a laser, the more energy is generated and the higher the ratio of absorption into an object becomes. Optimum materials for marking/processing depend on the wavelength of the laser beam. The following section explains the characteristics of each wavelength.

* Fibre laser marker: 1090 nm

Wavelength: 10600 nm | CO_2 laser marker



CO_2 lasers have a wavelength that is 10 times longer than a YAG, YVO_4 or Fibre laser. This is the longest wavelength among widely used industrial lasers. CO_2 lasers, as the name implies generate the laser medium through stimulation of CO_2 gas.

Typical characteristics of 10600 nm wavelength range lasers

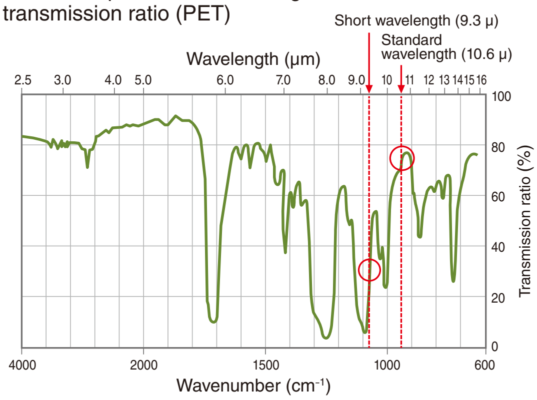
- Not absorbed well by metals
- Melting and burning occur due to the long wavelength and transfer of heat.
- Processing transparent objects such as glass and PET are possible.
- Contrast printing and discolouration are generally not possible with a CO_2 laser.

PET bottle

Polycarbonate

Sharp marking achieved with a CO_2 short-wavelength laser (9300 nm)

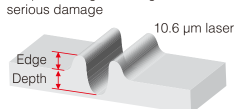
Relationship between wavelength and transmission ratio (PET)



CO_2 lasers achieve sharp marking/processing with less thermal influence when used with a shorter wavelength. As shown in the graph on the left, when the material is PET, nearly 80% of the laser with the standard wavelength (10.6 μm) passes through the target whereas the transmission ratio of a short wavelength (9.3 μm) is about 30%. Low transmission ratio means high absorption ratio. When the absorption ratio is high, the laser can efficiently raise the temperature of the marking surface and suppress thermal influence that might melt the area other than the surface layer, which enables clear and sharp marking.

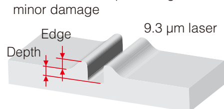
Standard wavelength

Deep and rough marking with serious damage



Short wavelength

Shallow and sharp marking with minor damage



Wavelength: 1064 nm | YVO$_4$/Fibre laser marker

The IR wavelength which is an abbreviation for Infrared Ray is the most versatile wavelength of light for laser processing. As the name implies, IR is the wavelength just greater than that of the red end of the visible spectrum, meaning it is invisible to the human eye (i.e. longer than 780 nm).

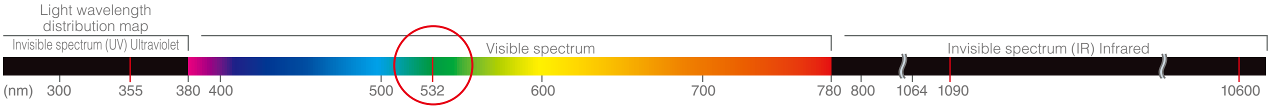
Typical characteristics of 1064 nm wavelength range lasers

- A wide range of processing applications from resins to metals
- Cannot process transparent objects like glass as the laser passes through such objects.
- Creates contrast on resins easily.

Moulded package

Automobile instrument panel component

Wavelength: 532 nm | Green laser marker

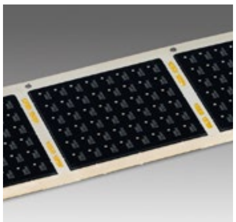


Green laser markers – also known as SHG (Second Harmonic Generation) laser markers – use 532 nm light to mark and process parts. The 532 nm wavelength is visible to humans and produced by passing 1064 nm light through an oxide single crystal (LBO: lithium borate).

Typical characteristics of 532 nm wavelength range lasers

- High absorption rates in materials that do not react well with typical IR wavelengths and those that reflect IR light such as gold and copper.
- Intricate processing is possible because of a smaller beam spot than IR lasers.
- Transparent objects are typically not able to be processed.

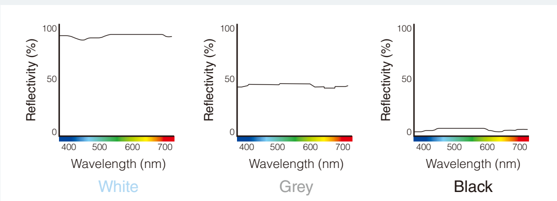
Low-profile package



Polyimide film

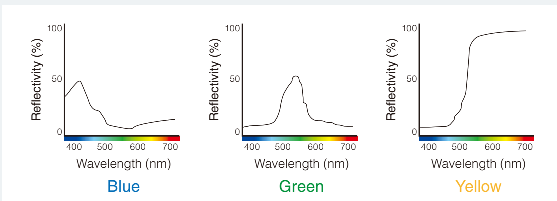
Influence on material depending on colour

Difference made by intensity



The graphs above show the reflectance of a laser depending on the intensity of the target. The reflectance for white, grey and black are shown as similar shapes, almost horizontal lines.
This means that a laser reflects in the same way regardless of its wavelength. Note, however, that the reflectance increases in the order of black, grey, and white. This shows that the darker the colour of the target is, the more a laser is absorbed.

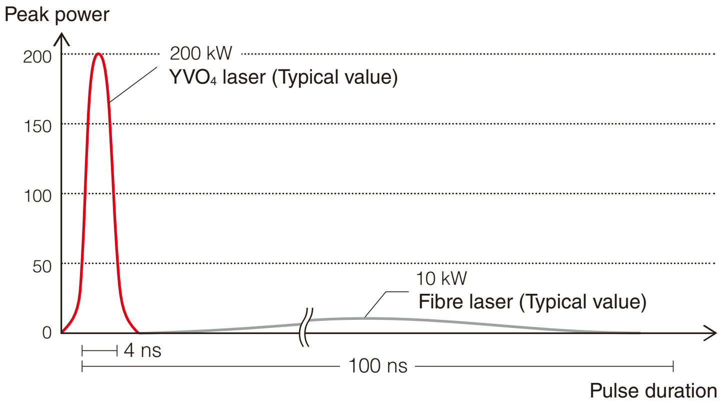
Difference made by colour



The graphs above show the reflectance of a laser depending on the colour of the target. The peaks of the reflectance for blue, green, and yellow come when the wavelength is 400 to 500 nm, 500 to 600 nm, and 500 nm or more respectively. This means that the absorption ratio of a laser decreases when a laser with a wavelength of the highest reflectance is used for marking.

Differences between YVO_4 and Fibre Lasers

Peak power and pulse duration

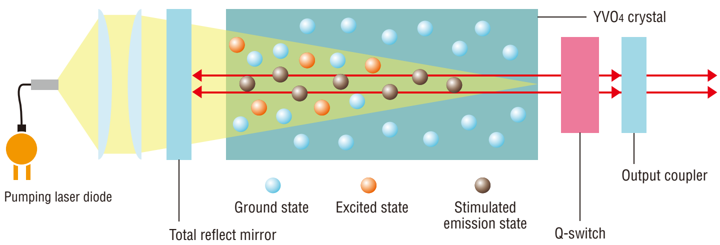


Output characteristics of YVO_4 and fibre lasers

The output characteristics of a laser vary considerably depending on laser excitation media. The major differences between YVO_4 and fibre lasers are the peak power and pulse duration. The peak power is the light intensity, and the pulse duration is the duration of illumination. YVO_4 lasers are produced by an excitation method based on end-pumping, which allows easy focusing of the laser beam spot and creates a high quality laser beam with high peak power and short pulses. On the other hand, fibre lasers have a high power output and low peak power compared with YVO_4 lasers.

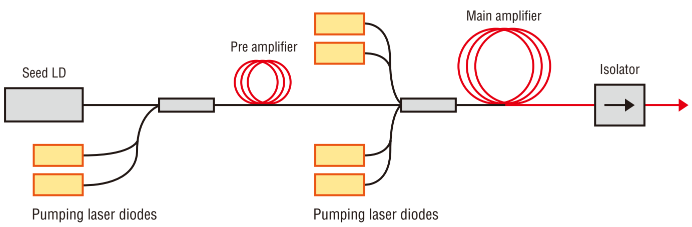
Differences in principle between YVO_4 and fibre lasers

YVO_4 lasers (End-pumping method)



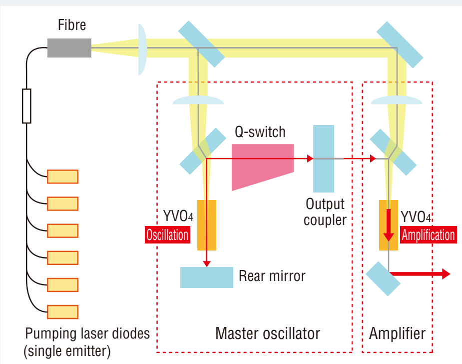
End-pumping method YVO_4 lasers are solid state lasers that use YVO_4 crystals for their laser medium. YVO_4 is a yttrium vanadate crystal and is doped with Nd (neodymium) in the same way as YAG lasers. In this method, pumped light is applied from one end of the YVO_4 crystal. The resonator is composed of a pair of mirrors with the crystal and Q-switch positioned between them.
Also, the degree of amplification at the centre of the crystal is large and the generated laser light is single mode, which makes it possible to output high-quality lasers.

Fibre laser



Fibre lasers use fibres as their medium and have long-distance communication relay amplification technology to maintain consistent high output. These optical fibres are composed of a core that propagates the light along the centre of the fibre and cladding that concentrically wraps the core. Fibre lasers use this core as the laser medium to amplify the light. The core is doped with Yb (ytterbium).

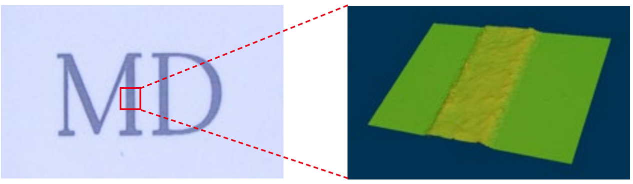
Hybrid laser (KEYENCE original)



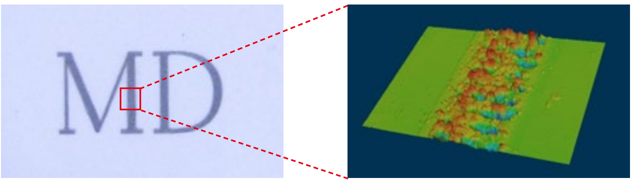
The KEYENCE Hybrid Laser Marker uses a next-generation laser oscillator that combines the high quality and high intensity of YVO_4 lasers with the long service life and excellent radiation characteristics of fibre lasers. One unique characteristic of the Hybrid Laser Marker is its two-stage construction in which a YVO_4 laser oscillator (master oscillator) is used to generate the pulse, which is then amplified by a YVO_4 amplifier. This makes it possible to amplify the pulse generated by the master oscillator while maintaining the high peak power and high quality of the pulse. Also, single-emitter pumping laser diodes—an advantage of fibre lasers—are used, which provide lower heat density than the multi-emitter laser diodes (laser diodes that have multiple light emitting surfaces in a single semiconductor chip) of solid state lasers. This enables the KEYENCE Hybrid Laser Marker to have a long service life even though it is a solid state laser.

Difference in marking quality between a YVO_4 laser and a fibre laser

▮KEYENCE YVO_4 laser marker



▮KEYENCE fibre laser marker



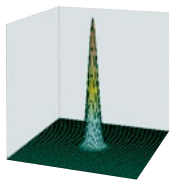
* The result may vary depending on the marking condition and material.

YVO_4 laser for high quality marking

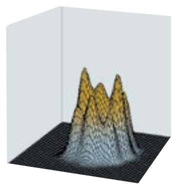
The figure on the left is a comparison of marking quality between a YVO_4 laser and a fibre laser. A YVO_4 laser produces a laser beam with high peak power and short pulses. Its ideal intensity distribution and short radiation time enable uniform, high quality marking with less thermal influence.* On the contrary, a fibre laser produces a laser beam with low peak power and long pulses. Its random intensity distribution and long radiation time increase thermal influence, which makes uniform marking difficult.

* When the end pumping method is used

Comparison of beam power distribution



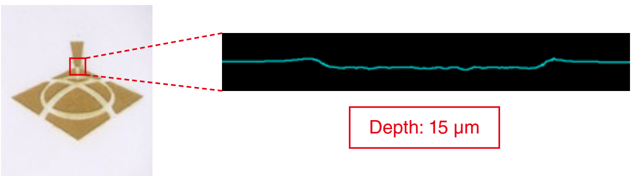
YVO_4 laser
(End pumping method)



Fibre laser

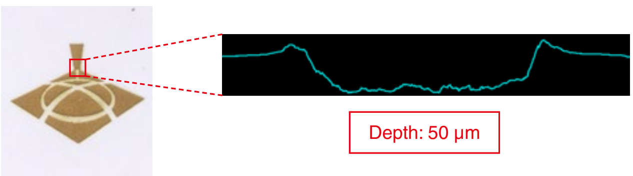
Differences in deep engraving between YVO_4 and fibre lasers

▮KEYENCE YVO_4 laser marker



Depth: 15 μm

▮KEYENCE fibre laser marker



Depth: 50 μm

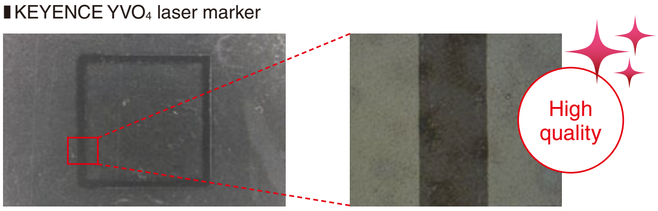
* The result may vary depending on the marking condition and material.

Fibre laser superior in carving and deep engraving

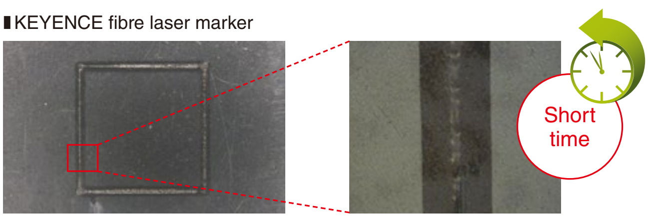
When the depth of engraving within the same marking time is compared for a resin target, a fibre laser engraved a depth of 50 μm and a YVO_4 laser engraved 15 μm. The pulse of a YVO_4 laser emits a high intensity laser beam to the material for a short time. The shallow engraved area in the surface layer is heated quickly and then cools immediately. Since the laser radiation stops before the heat is transferred, the thermal influence on the surrounding area is small. On the other hand, the pulse of a fibre laser emits a low intensity laser beam for a long time. The temperature of the material rises slowly and a liquefied or evaporating state continues for a long time. Consequently, a fibre laser is suitable for a large amount of engraving.

Differences in welding processing between YVO_4 and fibre lasers

▮KEYENCE YVO_4 laser marker



▮KEYENCE fibre laser marker

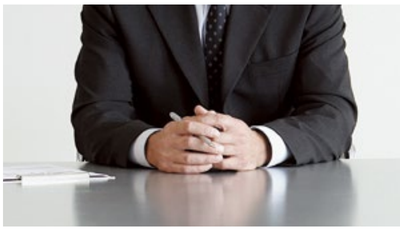


* The result may vary depending on the marking condition and material.

Continuous wave (CW) oscillation for constant heat application (YVO_4 laser)
High power output laser (Fibre laser)

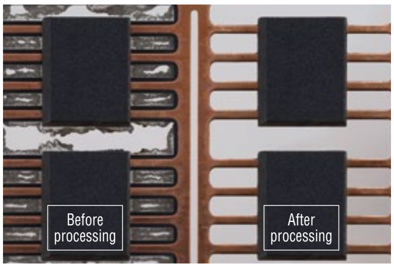
YVO_4 lasers allow selection of continuous wave oscillation or pulsed oscillation. Since welding requires uniform heat propagation, selecting continuous wave oscillation enables high quality welding. Fibre lasers produce a high power output and a long pulse laser that can apply heat intensely to complete welding within a short time.

LOCAL, IN-HOUSE TESTING LABS



Our customers have access to these test services provided by our dedicated sales engineers. To request a test, visit the KEYENCE website or contact your nearest KEYENCE office.

MARKING



PROCESSING

GLOBAL NETWORK CONTACT YOUR NEAREST OFFICE FOR RELEASE STATUS

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BELGIUM	FRANCE	INDONESIA	MEXICO	SINGAPORE	THAILAND
BRAZIL	GERMANY	ITALY	NETHERLANDS	SLOVAKIA	UK & IRELAND
CANADA	HONG KONG	JAPAN	PHILIPPINES	SLOVENIA	USA
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